

is mapped and the ratio of different types are generated. These operations are then repeated over and over to benchmark the processor. These traces cover tasks like basic spreadsheet and word processors, digital content creation, playing video games and intensive operations like 3D model rendering. Sometimes benchmarks will provide a time score which is basically how long the test took to finish. Here are a few benchmarks that will help you rate your processor.

PCMark

It's an industry standard benchmark that runs multiple tests and you can even check entire configurations of users on their leaderboards. This way you can even check configurations which are the exact same as yours and optimise your settings if your ranking is found to be lesser. PCMark even breaks down their benchmark suite into large verticals which are general usage scenarios for computers.

Certain applications make use of a single CPU thread while others are optimised to make use of all threads. PCMark does provide scores for both these usage scenarios. Also, 3DMark can be used to judge game physics scores which are often handled by the CPU.

Cinebench

This benchmark has two tests, one is a render test which gauges the CPU performance and the other is an OpenGL animation test. Cinebench is a good way



A pretty comprehensive benchmark

to estimate how your computer would fare if you are going to tinker with 3D rendering.

Blenchmark

On the same lines of Cinebench we also have Blenchmark which basically involves running a Blender test file and checking how long it takes to render a particular scene. Moreover, you can then check the leaderboards to see how well it fares against a plethora of other processors. <http://dgit.in/BlenchMark>

HOW TO TEST MORE

Our benchmark for CPUs cover all the aspects that we've already touched upon in the previous section and a few other benchmarks. We do run PCMark 8's entire series of tests which not only includes the creative suite but also the application tests that covers some commonly found applications like a spreadsheet, word processor and also a few applications from Adobe's creative suite. Then for video encoding, we use handbrake (a video encoder) that makes use of some of the more proprietary technologies like Quick Sync that actually speeds up the encoding process.

We then run a few gaming benchmarks to gauge the processor's performance in games that are CPU intensive, like some of Total War's offering – Rome 2 and the more recent Atilla. APU's and CPUs that have an Integrated Graphics Processor are tested on a lot more games and in certain cases like AMD's APU's, some of which can run in CrossFireX with AMD GPUs are configured as such and the performance increment is measured.

For 3D modelling, we use 3DS Max and Blender using a test file and the time consumed for the benchmark to complete is noted.

File compression is also benchmarked using WinRAR's built-in benchmark feature.

And lastly, we overclock the CPUs on air cooling and run SuperPi to figure out the most stable frequency that the CPU can handle and then repeat the entire set of benchmarks.



SuperPi

SuperPi is less functional in terms of letting the user know what tasks the processor is good at. It is mostly used to test stability of a processor once overclocked.

How can you say which is better?

Naming conventions for both brands are quite elaborate so here are a few links to help you understand what they are and what they mean. <http://dgit.in/IntelSuffix> <http://dgit.in/CPUNaming>. What you need to note here is that, there are plenty of CPU whose model numbers are exactly the same except for the suffix at the end. And the performance difference between the two are worlds apart.

For example, take the 4th Generation i5 Mobile processor that goes by the model number Core i5-4200. There are four variants with different suffixes, one is an H, then there an M, a U and lastly a Y. The H comes with the highest graphics offering possible for that architecture, the M simply stands for a dual/quad core mobile processor, the U is a low-powered variant for Ultrabooks and the Y takes it a step further and reduces power consumption a lot more than the U.

If you are surfing for a new laptop then you'll have noticed that most brick-and-mortar stores do not mention the entire model number and sometimes when they do so, the suffix is surreptitiously not mentioned. Just to put things into perspective, the individual rankings of these four pro-



Game physics are often handled by the CPU

cessors at the time of writing this article are as follows – 4200H: 394, 4200M: 440, 4200U: 594 and 4200Y: 819.

The more reliable (and obvious) method of telling which processor is better is by referring sites like passmark.com and notebookcheck.com which not only maintain leader boards but also display actual scores obtained.